

SOLIDWORKS API (NET VBA C#) — Reference

(93/337): IEdge

SOLIDWORKS API Documentation (converted from CHM)

IEdge (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IEdge Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEdge Interface

Allows access to its defining coedge, and adjacent faces, and its underlying curve and vertices as well as edge data.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IEdge
```

Visual Basic (Usage)



```
Dim instance As IEdge
```

C#

```
public interface IEdge
```

C++/CLI

```
public interface class IEdge
```

Visual Basic for Applications (VBA) Syntax

See [Edge](#).

Example

[Get Edge Curve Parameterization \(C#\)](#)
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[Edge](#)

See Also

[IEdge Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IEdge Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEdge Interface

The following tables list the members exposed by [IEdge](#).

Public Properties

Name	Description
Check	Gets whether the edge is a valid, and, if not, returns the faults.

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Public Methods

Name	Description
CreateWireBody	Creates a wire body from this edge.
Display	Highlights this edge with the specified color.
EdgeInFaceSense	Checks whether the edge and the loop lying on the specified face have the same direction (sense).
EnumCoEdges	Lists the coedges that reference this edge.
Evaluate	Obsolete. Superseded by IEdge.Evaluate2 .
Evaluate2	Evaluates the edge for the specified U parameter.
GetBody	Gets the body for this edge.
GetClosestPointOn	Uses the X,Y,Z input point and returns the closest point on the edge.
GetCoEdges	Gets the coedges that reference this edge.
GetCurve	Gets the underlying curve for this edge.
GetCurveParams	Obsolete. Superseded by IEdge.GetCurveParams2 .
GetCurveParams2	Obsolete. Superseded by IEdge.GetCurveParams3 .
GetCurveParams3	Gets a data object containing the curve parameters for this edge.
GetEndVertex	Gets the ending vertex for this edge.
GetID	Gets the edge ID of this edge in an imported body.
GetParameter	Gets the parameterization of the edge.
GetStartVertex	Gets the starting vertex for this edge.
GetTangentEdges	Gets all of the edges tangent to this edge.
GetTangentEdgesCount	Gets the number of edges tangent to this edge.
GetTrackingIDs	Gets the tracking IDs assigned to this edge .
GetTrackingIDsCount	Gets the number of tracking IDs on this edge.
GetTwoAdjacentFaces	Obsolete. Superseded by IEdge.GetTwoAdjacentFaces2 .
GetTwoAdjacentFaces2	Gets the two faces adjacent to an edge.
Highlight	Add highlights or removes highlights from this edge.
IEdgeInFaceSense	Obsolete. Superseded by IEdge.IEdgeInFaceSense2 .
IEdgeInFaceSense2	Checks whether the edge and the loop lying on the specified face have the same direction (sense).
IEvaluate	Obsolete. Superseded by IEdge.IEvaluate2 .
IEvaluate2	Evaluates the edge for the specified U parameter.
IGetClosestPointOn	Uses the X,Y,Z input point and returns the closest point on the edge.
IGetCurve	Gets the underlying curve for this edge.
IGetCurveParams	Obsolete. Superseded by IEdge.IGetCurveParams2 .

	IGetCurveParams2	Returns the curve parameters for this edge, including the edge and curve direction (sense).
	IGetEndVertex	Gets the ending vertex for this edge.
	IGetParameter	Gets the parameterization of the edge.
	IGetStartVertex	Gets the starting vertex for this edge.
	IGetTangentEdges	Gets all of the edges tangent to this edge.
	IGetTrackingIDs	Gets the tracking IDs assigned to this edge .
	IGetTwoAdjacentFaces	Obsolete. Superseded by IEdge::IGetTwoAdjacentFaces2
	IGetTwoAdjacentFaces2	Gets the two faces adjacent to an edge.
	IsParamReversed	Gets whether the edge and its underlying curve have the same parameterization or if the direction is reversed.
	IsTolerant	Gets whether an edge is tolerant and its tolerance value.
	RemoveId	Removes the edge ID assigned to this edge of an imported body.
	RemoveRedundantTopology	Removes redundant topology from the edge.
	RemoveTrackingID	Removes a tracking ID assigned to this edge .
	SetId	Sets the edge ID of this edge of an imported body.
	SetTrackingID	Assigns a tracking ID to this edge.

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See Also

[IEdge Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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IEdge Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEdge Interface

For a list of all members of this type, see [IEdge members](#).

Public Methods

Name	Description
 CreateWireBody	Creates a wire body from this edge.
 Display	Highlights this edge with the specified color.
 EdgeInFaceSense	Checks whether the edge and the loop lying on the specified face have the same direction (sense).
 EnumCoEdges	Lists the coedges that reference this edge.
 Evaluate	Obsolete. Superseded by IEdge::Evaluate2
 Evaluate2	Evaluates the edge for the specified U parameter.
 GetBody	Gets the body for this edge.
 GetClosestPointOn	Uses the X,Y,Z input point and returns the closest point on the edge.
 GetCoEdges	Gets the coedges that reference this edge.
 GetCurve	Gets the underlying curve for this edge.
 GetCurveParams	Obsolete. Superseded by IEdge::GetCurveParams2
 GetCurveParams2	Obsolete. Superseded by IEdge::GetCurveParams3
 GetCurveParams3	Gets a data object containing the curve parameters for this edge.
 GetEndVertex	Gets the ending vertex for this edge.
 GetId	Gets the edge ID of this edge in an imported body.
 GetParameter	Gets the parameterization of the edge.
 GetStartVertex	Gets the starting vertex for this edge.

		
	GetTangentEdges	Gets all of the edges tangent to this edge.
	GetTangentEdgesCount	Gets the number of edges tangent to this edge.
	GetTrackingIDs	Gets the tracking IDs assigned to this edge .
	GetTrackingIDsCount	Gets the number of tracking IDs on this edge.
	GetTwoAdjacentFaces	Obsolete. Superseded by IEdge::GetTwoAdjacentFaces2
	GetTwoAdjacentFaces2	Gets the two faces adjacent to an edge.
	Highlight	Add highlights or removes highlights from this edge.
	IEdgeInFaceSense	Obsolete. Superseded by IEdge::IEdgeInFaceSense2 .
	IEdgeInFaceSense2	Checks whether the edge and the loop lying on the specified face have the same direction (sense).
	IEvaluate	Obsolete. Superseded by IEdge::IEvaluate2 .
	IEvaluate2	Evaluates the edge for the specified U parameter.
	IGetClosestPointOn	Uses the X,Y,Z input point and returns the closest point on the edge.
	IGetCurve	Gets the underlying curve for this edge.
	IGetCurveParams	Obsolete. Superseded by IEdge::IGetCurveParams2 .
	IGetCurveParams2	Returns the curve parameters for this edge, including the edge and curve direction (sense).
	IGetEndVertex	Gets the ending vertex for this edge.
	IGetParameter	Gets the parameterization of the edge.
	IGetStartVertex	Gets the starting vertex for this edge.
	IGetTangentEdges	Gets all of the edges tangent to this edge.
	IGetTrackingIDs	Gets the tracking IDs assigned to this edge .
	IGetTwoAdjacentFaces	Obsolete. Superseded by IEdge::IGetTwoAdjacentFaces2 .
	IGetTwoAdjacentFaces2	Gets the two faces adjacent to an edge.
	IsParamReversed	Gets whether the edge and its underlying curve have the same parameterization or if the direction is reversed.
	IsTolerant	Gets whether an edge is tolerant and its tolerance value.
	RemoveId	Removes the edge ID assigned to this edge of an imported body.
	RemoveRedundantTopology	Removes redundant topology from the edge.
	RemoveTrackingID	Removes a tracking ID assigned to this edge .
	SetId	Sets the edge ID of this edge of an imported body.
	SetTrackingID	Assigns a tracking ID to this edge.

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 See Also

[IEdge Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IEdge Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEdge Interface

For a list of all members of this type, see [IEdge members](#).

 Public Properties

Name	Description
 Check	Gets whether the edge is a valid, and, if not, returns the faults.

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See Also

[IEdge Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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Check Property (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : Check Property (IEdge)

Gets whether the edge is a valid, and, if not, returns the faults.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Check As FaultEntity

Visual Basic (Usage)



```
Dim instance As IEdge
Dim value As FaultEntity

value = instance.Check
```

C#

```
FaultEntity Check {get;}
```

C++/CLI

```
property FaultEntity^ Check {
    FaultEntity^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pointer to [IFaultEntity](#) object

Visual Basic for Applications (VBA) Syntax

See [Edge::Check](#).

Example

[Check Edges for Faults \(C#\)](#)
[Check Edges for Faults \(VB.NET\)](#)
[Check Edges for Faults \(VBA\)](#)

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IFace2::Check Property](#)
[IBody2::Check3 Property](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12.4

SOLIDWORKS API Help

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CreateWireBody Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : CreateWireBody Method (IEdge)

Creates a wire body from this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateWireBody() As Body2
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As Body2
```

```
value = instance.CreateWireBody()
```

C#

```
Body2 CreateWireBody()
```

C++/CLI

```
Body2^ CreateWireBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the newly created [IBody2](#) object

Visual Basic for Applications (VBA) Syntax

See [Edge::CreateWireBody](#).

Remarks

Wire bodies contain wires, loops, coedges, edges, and vertices. Wires typically represent profiles, construction lines, and center lines of swept shapes. Wires can also represent wire frames that, when surfaced, form shells.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[ICurve::CreateWireBody Method](#)
[ImportedCurveFeatureData::SetBody Method](#)
[IModeler::CreateWireBody Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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Display Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : Display Method (IEdge)

Highlights this edge with the specified color.

.NET Syntax

Visual Basic (Declaration)

```
Sub Display(  
    ByVal Width As System.Integer, _  
    ByVal Red As System.Double, _  
    ByVal Green As System.Double, _  
    ByVal Blue As System.Double, _  
    ByVal HighlightState As System.Boolean _  
)
```

Visual Basic (Usage)

```
Dim instance As IEdge  
Dim Width As System.Integer  
Dim Red As System.Double  
Dim Green As System.Double  
Dim Blue As System.Double  
Dim HighlightState As System.Boolean  
  
instance.Display(Width, Red, Green, Blue, HighlightState)
```

C#

```
void Display(  
    System.int Width,  
    System.double Red,  
    System.double Green,  
    System.double Blue,  
    System.bool HighlightState  
)
```

C++/CLI

```
void Display(  
& System.int Width,  
& System.double Red,  
& System.double Green,  
& System.double Blue,  
& System.bool HighlightState  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Width

Highlight width

Red

Red value of RGB value for the color, between 0 and 1

Green

Green value of RGB value for the color, between 0 and 1

Blue

Blue value if RGB value for the color, between 0 and 1

HighlightState

True if the edge is highlighted, false if not

Visual Basic for Applications (VBA) Syntax

See [Edge::Display](#).

Example

[Add Highlighting to or Remove Highlighting From Edges \(VBA\)](#)
[Get Faces Affected by Feature \(VBA\)](#)

Remarks

To show the same edge with a different color, hide it and then set a different color. SOLIDWORKS shows the edge in the specified color until you hide it. Rotation, zoom, other repaint actions do not cause the edge to disappear.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IFace2::IHighlight Method](#)
[IFace2::IHighlight Method](#)
[IVertex::Display Method](#)
[IEdge::Highlight Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

EdgeInFaceSense Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : EdgeInFaceSense Method (IEdge)

Checks whether the edge and the loop lying on the specified face have the same direction (sense).

.NET Syntax

Visual Basic (Declaration)

```
Function EdgeInFaceSense( _  
    ByVal Facedisp As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim Facedisp As System.Object  
Dim value As System.Boolean
```

```
value = instance.EdgeInFaceSense(Facedisp)
```

C#

```
System.bool EdgeInFaceSense(  
    System.object Facedisp  
)
```

C++/CLI

```
System.bool EdgeInFaceSense(  
& System.Object^ Facedisp  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Facedisp
Face that the edge is on

Return Value

True for the same direction, false for the opposite direction

Visual Basic for Applications (VBA) Syntax

See [Edge::EdgeInFaceSense](#).

Remarks

If this edge does not belong to the specified face, then this method can cause unpredictable results.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::IEdgeInFaceSense2 Method](#)

SOLIDWORKS API Help

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EnumCoEdges Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : EnumCoEdges Method (IEdge)

Lists the coedges that reference this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function EnumCoEdges() As EnumCoEdges
```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim value As EnumCoEdges

value = instance.EnumCoEdges()
```

C#

```
EnumCoEdges EnumCoEdges()
```

C++/CLI

```
EnumCoEdges^ EnumCoEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [enumerated list of coedges](#)

Visual Basic for Applications (VBA) Syntax

See [Edge::EnumCoEdges](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEnumCoEdges::Next Method](#)

SOLIDWORKS API Help

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Evaluate Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : Evaluate Method (IEdge)

Obsolete. Superseded by [IEdge::Evaluate2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Evaluate( _
    ByVal Parameter As System.Double _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim Parameter As System.Double
Dim value As System.Object

value = instance.Evaluate(Parameter)
```

C#

```
System.object Evaluate(
    System.double Parameter
)
```

C++/CLI

```
System.Object^ Evaluate(
    & System.double Parameter
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameter

Value of the edge parameter

Return Value

Array values containing the x,y,z value and derivative of the edge (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Edge::Evaluate](#).

Example

[Select Tangent Edges Via Iteration \(VBA\)](#)

Remarks

Use [IEdge::GetCurveParams2](#) or [IEdge::IGetCurveParams2](#) to determine the valid parameter range for this method.

This OLE implementation of this method returns an array of doubles as follows:

[*PointX*, *PointY*, *PointZ*, *TangentX*, *TangentY*, *TangentZ*, *Success*]

where the point values are in meters and *Success* is True if successful and false if not.

The return value for the COM implementation is different. To determine success, check the HRESULT return value. The array is as follows:

[*PointX*, *PointY*, *PointZ*, *TangentX*, *TangentY*, *TangentZ*]

where the point values are specified in meters.

See Also

[IEdge Interface](#)

[IEdge Members](#)

[IEdge::IEvaluate Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Evaluate2 Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : Evaluate2 Method (IEdge)

Evaluates the edge for the specified U parameter.

.NET Syntax

Visual Basic (Declaration)

```
Function Evaluate2( _  
    ByVal Parameter As System.Double, _  
    ByVal NumberOfDerivatives As System.Integer _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IEdge  
Dim Parameter As System.Double  
Dim NumberOfDerivatives As System.Integer  
Dim value As System.Object  
  
value = instance.Evaluate2(Parameter, NumberOfDerivatives)
```

C#

```
System.object Evaluate2(  
    System.double Parameter,  
    System.int NumberOfDerivatives  
)
```

Copy Code

```
System.Object^ Evaluate2(
& System.double Parameter,
& System.int NumberOfDerivatives
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameter

Curve parameter U value (see **Remarks**)

NumberOfDerivatives

Number of derivatives (see **Remarks**)

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Edge::Evaluate2](#).

Example

1. Open a part in SOLIDWORKS.
2. Open an Immediate window.
3. Select an edge.
4. Run this VBA macro:

=====

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim boolstatus As BooleanDim longstatus As Long, longwarnings As Long
```

```
Type DoubleRec
    dValue As Double '64 bits, 8 bytes
End Type
```

```
Type Int2Rec
    iLower As Long '4 bytes
    iUpper As Long '4 bytes
End Type
```

```
' Unpack the status double
Function DoubleToLong(ByVal dVal As Double) As Long
    Dim dr As DoubleRec
    Dim i2r As Int2Rec
    ' Set the double value
    dr.dValue = dVal
    ' Copy the values
    LSet i2r = dr
    ' Extract the status value
    DoubleToLong = i2r.iLower
End Function
```

```
Sub main()

    Dim status As Long
    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc
    Dim myModelView As SldWorks.ModelView
    Set myModelView = Part.ActiveView
    myModelView.FrameState = swWindowState_e.swWindowMaximized
    Dim swSelectMgr As SldWorks.SelectionMgr
    Set swSelectMgr = Part.SelectionManager
    Dim swEdge As SldWorks.Edge
    Set swEdge = swSelectMgr.GetSelectedObject6(1, -1)
    Dim swCurve As SldWorks.Curve
    Dim swCurveParamData As SldWorks.CurveParamData
    Set swCurve = swEdge.GetCurve
```

```

Set swCurveParamData = swEdge.GetCurveParams3
Debug.Print "The maximum U parameter value is: " & swCurveParamData.UMaxValue
Debug.Print "The minimum U parameter value is: " & swCurveParamData.UMinValue
Dim pt As Variant
pt = swEdge.Evaluate2(swCurveParamData.UMaxValue, 0)
status = DoubleToLong(pt(UBound(pt)))
If status Then
    Debug.Print " The evaluated point at UMax is:  = " & pt(0) & ", " & pt(1) & ", " & pt(2)
Else
    Debug.Print "***** Error!"
End If

End Sub
=====

```

Remarks

Use [IEdge::GetCurveParams3](#) to determine the valid range of U parameter values for Parameter.

The returned array contains ((NumberOfDerivatives + 1) * 3) + 1 doubles:

[*evaluated_point*], [*evaluated_derivative_1*],...[*evaluated_derivative_NumberOfDerivatives*], [*status_code*]

where *status_code* is a packed double. After unpacking *status_code* into two longs, the lower long value is 1 for success or 0 for error. See the Example.

If curve type [ICurve::Identity](#) is...

Then the maximum number of derivatives is...

Line/circle/ellipse

2

Intersection curve

2

Constant parameter line

Determined by underlying surface

SP-curve

2

B-curve

Any number

For a curve of type swCurveTypes_e::TRIMMED_TYPE, the number of derivatives is determined by the base curve as obtained from [ICurve::GetBaseCurve](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBody Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetBody Method (IEdge)

Gets the body for this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBody() As Body2
```

Visual Basic (Usage)

```
Dim instance As IEdge
Dim value As Body2
```

```
value = instance.GetBody()
```

C#



Body2 GetBody()

C++/CLI

```
Body2^ GetBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IBody2](#) object for this edge

Visual Basic for Applications (VBA) Syntax

See [Edge::GetBody](#).

Example

[Extend Surface \(VBA\)](#)
[Create Temporary Bodies by Offsetting Surface Body \(C#\)](#)
[Create Temporary Bodies by Offsetting Surface Body \(VB.NET\)](#)
[Create Temporary Bodies by Offsetting Surface Body \(VBA\)](#)

See Also

[IEdge Interface](#)
[IEdge Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetClosestPointOn Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetClosestPointOn Method (IEdge)

Uses the X,Y,Z input point and returns the closest point on the edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetClosestPointOn( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IEdge  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.GetClosestPointOn(X, Y, Z)
```

C#

```
System.object GetClosestPointOn(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI



```
System.Object^ GetClosestPointOn(
& System.double X,
& System.double Y,
& System.double Z
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X value of the input point
Y
Y value of the input point
Z
Z value of the input point

Return Value

Array of five doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Edge::GetClosestPointOn](#).

Remarks

This method returns only one point, regardless of how many points achieve the minimum distance.

This method returns an array that contains the following values:

[*PointX*, *PointY*, *PointZ*, *UParameter*, *NotUsed*]

where the point values represent the point on the edge, and *UParameter* is the parameter on the edge that is closest to the input point. Ignore the fifth array element.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::IGetClosestPointOn Method](#)
[ICurve::GetClosestPointOn Method](#)
[ICurve::IGetClosestPointOn Method](#)
[IFace2::GetClosestPointOn Method](#)
[IFace2::IGetClosestPointOn Method](#)
[ISurface::GetClosestPointOn Method](#)
[ISurface::IGetClosestPointOn Method](#)
[IVertex::GetClosestPointOn Method](#)
[IVertex::IGetClosestPointOn Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCoEdges Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetCoEdges Method (IEdge)

Gets the coedges that reference this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCoEdges() As System.Object
```

Visual Basic (Usage)

```
Dim instance As IEdge
Dim value As System.Object

value = instance.GetCoEdges()
```

C#

```
System.object GetCoEdges()
```

C++/CLI



```
System.Object^ GetCoEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of coedges that reference the edge

Visual Basic for Applications (VBA) Syntax

See [Edge::GetCoEdges](#).

Example

[Select Loop of Edges \(VBA\)](#)
[Select Tangent Edges Via Iteration \(VBA\)](#)

Remarks

Call this method for body-related edges, not reference curve or sketch edges.

See Also

[IEdge Interface](#)
[IEdge Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCurve Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetCurve Method (IEdge)

Gets the underlying curve for this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCurve() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Object  
  
value = instance.GetCurve()
```

C#

```
System.object GetCurve()
```

C++/CLI

```
System.Object^ GetCurve();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the underlying [curve](#) for this edge

Visual Basic for Applications (VBA) Syntax

See [Edge::GetCurve](#).

Example

[Dimension Edge in Drawing \(VBA\)](#)
[Get Edge Curve Parameterization \(VBA\)](#)
[Get Entities Attached to Cosmetic Thread \(VBA\)](#)
[Get Intersecting Face and Edge \(VBA\)](#)
[Get Length of Edge \(VBA\)](#)
[Select Tangent Edges Via Iteration \(VBA\)](#)
[Fill Holes in Part \(C#\)](#)
[Fill Holes in Part \(VB.NET\)](#)
[Fill Holes in Part \(VBA\)](#)
[Select Edges of All Holes on Face \(C#\)](#)
[Select Edges of All Holes on Face \(VB.NET\)](#)
[Select Edges of All Holes on Face \(VBA\)](#)

See Also

[IEdge Interface](#)
[IEdge Members](#)
[ICoEdge::GetEdge Method](#)
[ICoEdge::GetCurve Method](#)
[IEdge::GetCurveParams2 Method](#)
[ICoEdge::GetCurveParams Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCurveParams Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetCurveParams Method (IEdge)

Obsolete. Superseded by [IEdge::GetCurveParams2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetCurveParams() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim value As System.Object

value = instance.GetCurveParams()
```

C#

```
System.object GetCurveParams()
```

C++/CLI

```
System.Object^ GetCurveParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Edge::GetCurveParams](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCurveParams2 Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetCurveParams2 Method (IEdge)

Obsolete. Superseded by [IEdge::GetCurveParams3](#).

.NET Syntax

```
Function GetCurveParams2() As System.Object
```



```
Dim instance As IEdge
Dim value As System.Object

value = instance.GetCurveParams2()
```

```
System.object GetCurveParams2()
```

```
System.Object^ GetCurveParams2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array values describing the parameterization of the edge

Visual Basic for Applications (VBA) Syntax

See [Edge::GetCurveParams2](#).

Example

[Dimension Edge in Drawing \(VBA\)](#)
[Get Edge Curve Parameterization \(VBA\)](#)
[Get Intersecting Face and Edge \(VBA\)](#)
[Get Length of Edge \(VBA\)](#)
[Select Tangent Edges Via Iteration \(VBA\)](#)

Remarks

Calls to this method must be preceded by a call to [IEdge::GetCurve](#) or [IEdge::IGetCurve](#) because SOLIDWORKS does not keep the underlying curve information. [Edge::GetCurve](#) generates this information so that you can extract the curve parameters.

You can use the data returned by this method to determine if a circular edge is a complete circle or an arc.

The return value is the following array of 11 doubles:

[*StartPtX*, *StartPtY*, *StartPtZ*, *EndPtX*, *EndPtY*, *EndPtZ*, *StartUParam*, *EndUParam*, *PackDouble1*, *PackDouble2*, *PackDouble3*]

where *PackDouble1*, *PackDouble2*, and *PackDouble3* are each a set of two integers packed into a double. These variables hold the following pair of integers:

PackDouble1	Two packed integers: 1. Unused 2. <i>curveType</i> as documented in ICurve::Identity
PackDouble2	Two packed integers: 1. Unused 2. <i>curveTag</i>
PackDouble3	Two packed integers: 1. Unused 2. <i>SenseFlag</i> (indicates whether the curve and edge are in the same direction)

If *SenseFlag* is True, then the curve and edge are in the same direction. If *SenseFlag* is false, then the curve and edge are in opposite directions. In this case, this method returns the start parameter (point) that corresponds to the end of the edge and the end parameter (point) that corresponds to the end of the edge.

The start parameter is always smaller than the end parameter. If the curve and edge are in opposite directions, then the start and end parameters are along the negative portion of the parameter space. For example, if the start parameter of the edge is 10 and the end parameter is 5, then the parameter range returned is -10, -5. This ensures that the start parameter is smaller than the end parameter. To correct the values fo

the edge, swap the two values between the start and end parameters and then negate both values so that the value of the start parameter is 5 and the end parameter is 10.

If the curve is closed and the curve starts and ends at the same point, then *StartUParam* and *EndUParam* are a period apart.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::IGetCurveParams2 Method](#)
[ICoEdge::GetCurveParams Method](#)
[ICoEdge::IGetCurveParams Method](#)
[IEdge::IsParamReversed Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCurveParams3 Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetCurveParams3 Method (IEdge)

Gets a data object containing the curve parameters for this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCurveParams3() As CurveParamData
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As CurveParamData  
  
value = instance.GetCurveParams3()
```

C#

```
CurveParamData GetCurveParams3()
```

C++/CLI

```
CurveParamData^ GetCurveParams3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

An [ICurveParamData](#) object

Visual Basic for Applications (VBA) Syntax

See [Edge::GetCurveParams3](#).

Example

[Get Edge Curve Parameterization \(C#\)](#)
[Get Edge Curve Parameterization \(VB.NET\)](#)
[Get Edge Curve Parameterization \(VBA\)](#)

Remarks

Before calling this method, you must call [IEdge::GetCurve](#) to generate the underlying curve information.

You can use the data returned by this method to determine if a circular edge is a complete circle or an arc.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[ICoEdge::GetCurveParams Method](#)
[IEdge::IsParamReversed Method](#)
[ICurve::GetBCurveParams5 Method](#)
[ICurve::GetPCurveParams2 Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEndVertex Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetEndVertex Method (IEdge)

Gets the ending vertex for this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEndVertex() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim value As System.Object

value = instance.GetEndVertex()
```

C#

```
System.Object GetEndVertex()
```

C++/CLI

```
System.Object^ GetEndVertex();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Vertex

Visual Basic for Applications (VBA) Syntax

See [Edge::GetEndVertex](#).

Example

[Check Interference Using IModeler::CheckInterference \(VBA\)](#)
[Get Radius of Variable Radius Fillet \(VBA\)](#)
[Get Vertex \(VBA\)](#)
[Modify Fillet Weld Bead \(VBA\)](#)
[Modify Fillet Weld Bead \(VB.NET\)](#)
[Modify Fillet Weld Bead \(C#\)](#)

Remarks

If no vertex exists for this edge (such as the edge of a newly created cylinder), then this method returns NULL.

This method and [IEdge::GetStartVertex](#) or [IEdge::!GetStartVertex](#) return distinct vertices, unless the edge is closed. Because edges have no natural direction, you cannot necessarily predict which of the two points you will get first and which last.

Use [ICoEdge::GetCurveParams](#) or [ICoEdge::!GetCurveParams](#) to get consistent start and end positions.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::IGetEndVertex Method](#)
[IEdge::GetStartVertex Method](#)
[IEdge::IGetStartVertex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetID Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetID Method (IEdge)

Gets the edge ID of this edge in an imported body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetID() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Integer  
  
value = instance.GetID()
```

C#

```
System.int GetID()
```

C++/CLI

```
System.int GetID();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Edge ID of this edge in an imported body

Visual Basic for Applications (VBA) Syntax

See [Edge::GetID](#).

Remarks

SOLIDWORKS uses this ID to track edges of imported bodies.

The ID of the edge of an imported body:

- is not saved with the document.
- must be unique.
- can be changed by any third-party application. The intent is that if you [assign an ID to an edge of an imported body](#), you can refer to that edge within your application.
- is not the same as a [persistent reference ID](#) or [tracking ID](#).

Use the [IAttribute](#) object to store data with an edge.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::RemoveId Method](#)
[IEdge::SetId Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetParameter Method (IEdge)

[See Also](#)



Gets the parameterization of the edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetParameter( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As System.Object

value = instance.GetParameter(X, Y, Z)
```

C#

```
System.object GetParameter(
    System.double X,
    System.double Y,
    System.double Z
)
```

C++/CLI

```
System.Object^ GetParameter(
    & System.double X,
    & System.double Y,
    & System.double Z
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X value
Y
Y value
Z
Z value

Return Value

Array containing parameterization of the edge

Visual Basic for Applications (VBA) Syntax

See [Edge::GetParameter](#).

Remarks

This method returns an array of 2 doubles:

- retval[0] - Parametric value of the specified point
- retval[1] - BOOL value; True for success, false for failure

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::IGetParameter Method](#)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetStartVertex Method (IEdge)

Gets the starting vertex for this edge.

.NET Syntax

Visual Basic (Declaration)

Function GetStartVertex() As System.Object

Visual Basic (Usage)



```
Dim instance As IEdge
Dim value As System.Object

value = instance.GetStartVertex()
```

C#

```
System.object GetStartVertex()
```

C++/CLI

```
System.Object^ GetStartVertex();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Vertex

Visual Basic for Applications (VBA) Syntax

See [Edge::GetStartVertex](#).

Example

[Check Interference Using IModeler::CheckInterference \(VBA\)](#)
[Display Vertices \(VBA\)](#)
[Get Radius of Variable Radius Fillet \(VBA\)](#)
[Get Vertex \(VBA\)](#)
[Modify Fillet Weld Bead \(VBA\)](#)
[Modify Fillet Weld Bead \(VB.NET\)](#)
[Modify Fillet Weld Bead \(C#\)](#)

Remarks

If no vertex exists for this edge (such as the edge of a newly created cylinder), then this method returns null.

This method and [IEdge::GetEndVertex](#) or [IEdge::!GetEndVertex](#) return distinct vertices, unless the edge is closed. Because edges have no natural direction, you cannot necessarily predict which of the two points you will get first and which last.

Use [ICoEdge::GetCurveParams](#) or [ICoEdge::!GetCurveParams](#) to get consistent start and end positions.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::!GetStartVertex Method](#)
[IEdge::GetEndVertex Method](#)
[IEdge::!GetEndVertex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTangentEdges Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetTangentEdges Method (IEdge)

Gets all of the edges tangent to this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTangentEdges() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Object  
  
value = instance.GetTangentEdges()
```

C#

```
System.object GetTangentEdges()
```

C++/CLI

```
System.Object^ GetTangentEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of tangent edges

Visual Basic for Applications (VBA) Syntax

See [Edge::GetTangentEdges](#).

Example

[Select Tangent Edges Topologically \(VBA\)](#)

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::IGetTangentEdges Method](#)
[IEdge::GetTangentEdgesCount Method](#)
[IModelDoc2::SelectTangency Method](#)

Availability

SOLIDWORKS 2001Pus SP1, Revision Number 10.1

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTangentEdgesCount Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetTangentEdgesCount Method (IEdge)

Gets the number of edges tangent to this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTangentEdgesCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Integer  
  
value = instance.GetTangentEdgesCount()
```

C#

```
System.int GetTangentEdgesCount()
```

C++/CLI

```
System.int GetTangentEdgesCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of tangent edges

Visual Basic for Applications (VBA) Syntax

See [Edge::GetTangentEdgesCount](#).

Remarks

Call this method before calling [IEdge::GetTangentEdges](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetTangentEdges Method](#)

Availability

SOLIDWORKS 2001Plus SP1, Revision Number 10.1

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTrackingIDs Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetTrackingIDs Method (IEdge)

Gets the [tracking IDs assigned to this edge](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrackingIDs( _  
    ByVal TrackingCookie As System.Integer, _  
    ByRef TrackingIDs As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim TrackingCookie As System.Integer  
Dim TrackingIDs As System.Object  
Dim value As System.Integer  
  
value = instance.GetTrackingIDs(TrackingCookie, TrackingIDs)
```

C#

```
System.int GetTrackingIDs(  
    System.int TrackingCookie,  
    out System.object TrackingIDs  
)
```

C++/CLI

```
System.int GetTrackingIDs(  
& System.int TrackingCookie,  
& [Out] System.Object^ TrackingIDs  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie

Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

TrackingIDs

Array of tracking IDs on this edge

Return Value

Status as defined by [swTrackingIDError_e](#)

Visual Basic for Applications (VBA) Syntax

See [Edge::GetTrackingIDs](#).

Example

[Get, Set, and Remove Tracking IDs on Edge \(VBA\)](#)

Remarks

See [Tracking IDs](#) for more information.

See Also

[IEdge Interface](#)

[IEdge Members](#)

[IEdge::GetTrackingIDsCount Method](#)

[IEdge::IGetTrackingIDs Method](#)

[IEdge::RemoveTrackingID Method](#)

[IEdge::SetTrackingID Method](#)

[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTrackingIDsCount Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetTrackingIDsCount Method (IEdge)

Gets the number of tracking IDs on this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrackingIDsCount( _  
    ByVal TrackingCookie As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim TrackingCookie As System.Integer  
Dim value As System.Integer  
  
value = instance.GetTrackingIDsCount(TrackingCookie)
```

C#

```
System.int GetTrackingIDsCount(  
    System.int TrackingCookie  
)
```

C++/CLI

```
System.int GetTrackingIDsCount(  
    & System.int TrackingCookie
```

)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie

Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

Return Value

Number of tracking IDs on this edge

Visual Basic for Applications (VBA) Syntax

See [Edge::GetTrackingIDsCount](#).

Example

[Get, Set, and Remove Tracking IDs on Edge \(VBA\)](#)

Remarks

Call this method before calling [IEdge::!GetTrackingIDs](#) to determine the size for that array.

See [Tracking IDs](#) for more information.

See Also

[IEdge Interface](#)

[IEdge Members](#)

[IEdge::GetTrackingIDs Method](#)

[IEdge::!GetTrackingIDs Method](#)

[IEdge::RemoveTrackingID Method](#)

[IEdge::SetTrackingID Method](#)

[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTwoAdjacentFaces Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetTwoAdjacentFaces Method (IEdge)

Obsolete. Superseded by [IEdge::GetTwoAdjacentFaces2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetTwoAdjacentFaces() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Object
```

```
value = instance.GetTwoAdjacentFaces()
```

C#

```
System.object GetTwoAdjacentFaces()
```

C++/CLI

```
System.Object^ GetTwoAdjacentFaces();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Edge::GetTwoAdjacentFaces](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTwoAdjacentFaces2 Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : GetTwoAdjacentFaces2 Method (IEdge)

Gets the two faces adjacent to an edge.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTwoAdjacentFaces2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Object
```

```
value = instance.GetTwoAdjacentFaces2()
```

C#

```
System.Object GetTwoAdjacentFaces2()
```

C++/CLI

```
System.Object^ GetTwoAdjacentFaces2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing two adjacent [faces](#)

Visual Basic for Applications (VBA) Syntax

See [Edge::GetTwoAdjacentFaces2](#).

Example

[Get Angle of Hole Not Normal to a Face \(VBA\)](#)

Remarks

This method is designed to be used with body-related edges, not reference curve or reference sketch edges. This method supports both solid and sheet bodies.

If...	Then the values returned are...
Two valid faces exist	Two faces
Only one valid face exists	One face and NULL

See Also

[IEdge Interface](#)
[IEdge Members](#)

[IEdge::IGetTwoAdjacentFaces2 Method](#)
[IVertex::GetAdjacentFaces Method](#)
[IVertex::IGetAdjacentFaces Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Highlight Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : Highlight Method (IEdge)

Add highlights or removes highlights from this edge.

.NET Syntax

Visual Basic (Declaration)

```
Sub Highlight( _  
    ByVal State As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim State As System.Boolean  
  
instance.Highlight(State)
```

C#

```
void Highlight(  
    System.bool State  
)
```

C++/CLI

```
void Highlight(  
& System.bool State  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

State

True adds highlights to this edge, false removes highlights from this edge

Visual Basic for Applications (VBA) Syntax

See [Edge::Highlight](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::Display Method](#)
[IFace2::Highlight Method](#)
[IFace2::IHighlight Method](#)
[IVertex::Display Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IEdgeInFaceSense Method (IEdge)

[See Also](#)



Obsolete. Superseded by [IEdge::IEdgeInFaceSense2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IEdgeInFaceSense( _  
    ByVal Facedisp As Face _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim Facedisp As Face  
Dim value As System.Boolean  
  
value = instance.IEdgeInFaceSense(Facedisp)
```

C#

```
System.bool IEdgeInFaceSense(  
    Face Facedisp  
)
```

C++/CLI

```
System.bool IEdgeInFaceSense(  
& Face^ Facedisp  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Facedisp

Visual Basic for Applications (VBA) Syntax

See [Edge::IEdgeInFaceSense](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IEdgeInFaceSense2 Method (IEdge)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IEdge Interface](#) : IEdgeInFaceSense2 Method (IEdge)

Checks whether the edge and the loop lying on the specified face have the same direction (sense).

.NET Syntax

Visual Basic (Declaration)

```
Function IEdgeInFaceSense2( _  
    ByVal Facedisp As Face2 _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim Facedisp As Face2  
Dim value As System.Boolean
```

```
value = instance.IEdgeInFaceSense2(Facedisp)
```

C#

```
System.bool IEdgeInFaceSense2(  
    Face2 Facedisp  
)
```

C++/CLI

```
System.bool IEdgeInFaceSense2(  
& Face2^ Facedisp  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Facedisp

Pointer to the [face](#) that the edge is on

Return Value

True for same direction as the loop, false for opposite

Visual Basic for Applications (VBA) Syntax

See [Edge::IEdgeInFaceSense2](#).

Remarks

If this edge does not belong to the face that is passed as an argument, unpredictable results can occur.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::EdgeInFaceSense Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IEvaluate Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IEvaluate Method (IEdge)

Obsolete. Superseded by [IEdge::IEvaluate2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IEvaluate( _  
    ByVal Parameter As System.Double _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim Parameter As System.Double  
Dim value As System.Double  
  
value = instance.IEvaluate(Parameter)
```

C#

```
System.double IEvaluate(  
    System.double Parameter  
)
```

C++/CLI

```
System.double IEvaluate(  
& System.double Parameter  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameter

Value of the edge parameter

Return Value

Pointer to an array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Edge::IEvaluate](#).

Remarks

Use [IEdge::GetCurveParams2](#) or [IEdge::IGetCurveParams2](#) to determine the valid parameter range for this method.

This OLE implementation of this method returns an array of doubles as follows:

[*PointX*, *PointY*, *PointZ*, *TangentX*, *TangentY*, *TangentZ*, *Success*]

where the point values are in meters and *Success* is True if successful and false if not.

The return value for the COM implementation is different. To determine success, check the HRESULT return value. The array is as follows:

[*PointX*, *PointY*, *PointZ*, *TangentX*, *TangentY*, *TangentZ*]

where the point values are specified in meters.

See Also

[IEdge Interface](#)

[IEdge Members](#)

[IEdge::Evaluate Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IEvaluate2 Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IEvaluate2 Method (IEdge)

Evaluates the edge for the specified U parameter.

.NET Syntax

Visual Basic (Declaration)

```
Function IEvaluate2(  
    ByVal Parameter As System.Double, _  
    ByVal NumberOfDerivatives As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```

Dim instance As IEdge
Dim Parameter As System.Double
Dim NumberOfDerivatives As System.Integer
Dim value As System.Double

value = instance.IEvaluate2(Parameter, NumberOfDerivatives)

```

C#

```

System.double IEvaluate2(
    System.double Parameter,
    System.int NumberOfDerivatives
)

```

C++/CLI

```

System.double IEvaluate2(
    & System.double Parameter,
    & System.int NumberOfDerivatives
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameter
Curve parameter U value (see **Remarks**)

NumberOfDerivatives
Number of derivatives (see **Remarks**)

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Edge::IEvaluate2](#).

Remarks

Use [IEdge::IGetCurveParams2](#) to determine the valid range of U parameter values for Parameter.

The returned array contains ((NumberOfDerivatives + 1) * 3) + 1 doubles:

[*evaluated_point*], [*evaluated_derivative_1*],...[*evaluated_derivative_NumberOfDerivatives*], [*status_code*]

where *status_code* is a packed double. After unpacking *status_code* into two longs, the lower long value is 1 for success or 0 for error. See the Example in [IEdge::Evaluate2](#).

If curve type [ICurve::Identity](#) is...

Then the maximum number of derivatives is...

Line/circle/ellipse	2
Intersection curve	2
Constant parameter line	Determined by underlying surface
SP-curve	2
B-curve	Any number

For a curve of type swCurveTypes_e::TRIMMED_TYPE, the number of derivatives is determined by the base curve as obtained from [ICurve::GetBaseCurve](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetCurveParams2 Method](#)

Availability

SOLIDWORKS 2007 SP5, Revision Number 15.5

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetClosestPointOn Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IGetClosestPointOn Method (IEdge)

Uses the X,Y,Z input point and returns the closest point on the edge.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetClosestPointOn( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Double  
  
value = instance.IGetClosestPointOn(X, Y, Z)
```

C#

```
System.double IGetClosestPointOn(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.double IGetClosestPointOn(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X](#)
X value of the input point

[Y](#)
Y value of the input point

[Z](#)
Z value of the input point

Return Value

Pointer to an array of five doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Edge.IGetClosestPointOn](#).

Remarks

This method returns only one point, regardless of how many points achieve the minimum distance.

This method returns an array that contains the following values:

[*PointX*, *PointY*, *PointZ*, *UParameter*, *NotUsed*]

where the point values represent the point on the edge, and *UParameter* is the parameter on the edge that is closest to the input point. Ignore the fifth array element.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetClosestPointOn Method](#)
[ICurve::GetClosestPointOn Method](#)
[ICurve::IGetClosestPointOn Method](#)
[IFace2::GetClosestPointOn Method](#)
[IFace2::IGetClosestPointOn Method](#)
[ISurface::GetClosestPointOn Method](#)
[ISurface::IGetClosestPointOn Method](#)
[IVertex::GetClosestPointOn Method](#)
[IVertex::IGetClosestPointOn Method](#)

SOLIDWORKS API Help

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IGetCurve Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IGetCurve Method (IEdge)

Gets the underlying curve for this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetCurve() As Curve
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As Curve  
  
value = instance.IGetCurve()
```

C#

```
Curve IGetCurve()
```

C++/CLI

```
Curve^ IGetCurve();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the underlying [curve](#) for this edge

Visual Basic for Applications (VBA) Syntax

See [Edge::IGetCurve](#).

Example

[Get Circular Holes in Face \(C++\)](#)

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetCurve Method](#)
[ICoEdge::GetCurve Method](#)
[ICoEdge::IGetCurve Method](#)
[ICoEdge::GetEdge Method](#)
[ICoEdge::IGetEdge Method](#)
[ICoEdge::GetCurveParams Method](#)
[ICoEdge::IGetCurveParams Method](#)

IGetCurveParams Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IGetCurveParams Method (IEdge)

Obsolete. Superseded by [IEdge::IGetCurveParams2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetCurveParams() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Double
```

```
value = instance.IGetCurveParams()
```

C#

```
System.double IGetCurveParams()
```

C++/CLI

```
System.double IGetCurveParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Edge::IGetCurveParams](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)

IGetCurveParams2 Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IGetCurveParams2 Method (IEdge)

Returns the curve parameters for this edge, including the edge and curve direction (sense).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetCurveParams2() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Double
```

```
value = instance.IGetCurveParams2()
```

C#

```
System.double IGetCurveParams2()
```

C++/CLI

```
System.double IGetCurveParams2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Example

[Get Circular Holes in Face \(C++\)](#)

Remarks

Calls to this method must be preceded by a call to [IEdge::GetCurve](#) or [IEdge::IGetCurve](#) because SOLIDWORKS does not keep the underlying curve information. [Edge::GetCurve](#) generates this information so that you can extract the curve parameters.

You can use the data returned by this method to determine if a circular edge is a complete circle or an arc.

The return value is the following array of 11 doubles:

[*StartPtX*, *StartPtY*, *StartPtZ*, *EndPtX*, *EndPtY*, *EndPtZ*, *StartUParam*, *EndUParam*, *PackDouble1*, *PackDouble2*, *PackDouble3*]

where *PackDouble1*, *PackDouble2*, and *PackDouble3* are each a set of two integers packed into a double. These variables hold the following pair of integers:

PackDouble1	Two packed integers: 1. Unused 2. curveType as documented in ICurve::Identity
PackDouble2	Two packed integers: 1. Unused 2. curveTag
PackDouble3	Two packed integers: 1. Unused 2. SenseFlag (indicates whether the curve and edge are in the same direction)

If SenseFlag is True, then the curve and edge are in the same direction. If SenseFlag is false, then the curve and edge are in opposite directions. In this case, this method returns the start parameter (point) that corresponds to the end of the edge and the end parameter (point) that corresponds to the end of the edge.

The start parameter is always smaller than the end parameter. If the curve and edge are in opposite directions, then the start and end parameters are along the negative portion of the parameter space. For example, if the start parameter of the edge is 10 and the end parameter is 5, then the parameter range returned is -10, -5. This ensures that the start parameter is smaller than the end parameter. To correct the values for the edge, swap the two values between the start and end parameters and then negate both values so that the value of the start parameter is 5 and the end parameter is 10.

If the curve is closed and the curve starts and ends at the same point, then *StartUParam* and *EndUParam* are a period apart.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetCurveParams2 Method](#)
[ICoEdge::GetCurveParams Method](#)
[ICoEdge::IGetCurveParams Method](#)
[IEdge::IsParamReversed Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetEndVertex() As Vertex
```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim value As Vertex

value = instance.IGetEndVertex()
```

C#

```
Vertex IGetEndVertex()
```

C++/CLI

```
Vertex^ IGetEndVertex();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [vertex](#)

 Visual Basic for Applications (VBA) Syntax

See [Edge::IGetEndVertex](#).

 Remarks

If no vertex exists for this edge (such as the edge of a newly created cylinder), then this method returns NULL.

This method and [IEdge::GetStartVertex](#) or [IEdge::IGetStartVertex](#) return distinct vertices, unless the edge is closed. Because edges have no natural direction, you cannot necessarily predict which of the two points you will get first and which last.

Use [ICoEdge::GetCurveParams](#) or [ICoEdge::IGetCurveParams](#) to get consistent start and end positions.

 See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetEndVertex Method](#)
[IEdge::GetStartVertex Method](#)
[IEdge::IGetStartVertex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetParameter Method (IEdge)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > IEdge Interface : IGetParameter Method (IEdge)

Gets the parameterization of the edge.

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetParameter( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As System.Double
```

Visual Basic (Usage)



```

Dim instance As IEdge
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As System.Double

value = instance.IGetParameter(X, Y, Z)

```

C#

```

System.double IGetParameter(
    System.double X,
    System.double Y,
    System.double Z
)

```

C++/CLI

```

System.double IGetParameter(
    & System.double X,
    & System.double Y,
    & System.double Z
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X value

Y
Y value

Z
Z value

Return Value

Pointer to an array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Edge::IGetParameter](#).

Remarks

This method returns an array of 2 doubles:

- retVal[0] - Parametric value of the specified point
- retVal[1] - BOOL value; True for success, false for failure

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetParameter Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetStartVertex Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IGetStartVertex Method (IEdge)

Gets the starting vertex for this edge.

.NET Syntax

Visual Basic (Declaration)

```

Function IGetStartVertex() As Vertex

```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim value As Vertex

value = instance.IGetStartVertex()
```

C#

```
Vertex IGetStartVertex()
```

C++/CLI

```
Vertex^ IGetStartVertex();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [vertex](#)

Visual Basic for Applications (VBA) Syntax

See [Edge::IGetStartVertex](#).

Remarks

If no vertex exists for this edge (such as the edge of a newly created cylinder), then this method returns NULL.

This method and [IEdge::GetEndVertex](#) or [IEdge::IGetEndVertex](#) return distinct vertices, unless the edge is closed. Because edges have no natural direction, you cannot necessarily predict which of the two points you will get first and which last.

Use [ICoEdge::GetCurveParams](#) or [ICoEdge::IGetCurveParams](#) to get consistent start and end positions.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetStartVertex Method](#)
[IEdge::GetEndVertex Method](#)
[IEdge::IGetEndVertex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTangentEdges Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IGetTangentEdges Method (IEdge)

Gets all of the edges tangent to this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTangentEdges( _
    ByVal Count As System.Integer _
) As Edge
```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim Count As System.Integer
Dim value As Edge

value = instance.IGetTangentEdges(Count)
```

C#

```
Edge IGetTangentEdges(
    System.int Count
)
```

```
Edge^ IGetTangentEdges(
& System.int Count
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of tangent edges

Return Value

- in-process, unmanaged C++: Pointer to an array of [edges](#) of size Count
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [IEdge::GetTangentEdgesCount](#) before this method.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetTangentEdges Method](#)
[IModelDoc2::SelectTangency Method](#)

Availability

SOLIDWORKS 2001Plus SP1, Revision Number 10.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTrackingIDs Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IGetTrackingIDs Method (IEdge)

Gets the [tracking IDs assigned to this edge](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTrackingIDs( _
    ByVal TrackingCookie As System.Integer, _
    ByVal Count As System.Integer, _
    ByRef TrackingIDs As System.Integer _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim TrackingCookie As System.Integer
Dim Count As System.Integer
Dim TrackingIDs As System.Integer
Dim value As System.Integer
```

```
value = instance.IGetTrackingIDs(TrackingCookie, Count, TrackingIDs)
```

C#

```
System.int IGetTrackingIDs(
    System.int TrackingCookie,
    System.int Count,
    out System.int TrackingIDs
)
```

```
System.int IGetTrackingIDs(  
& System.int TrackingCookie,  
& System.int Count,  
& [Out] System.int TrackingIDs  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie

Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

Count

Number of tracking IDs on this edge

TrackingIDs

- in-process, unmanaged C++: Pointer to an array of tracking IDs on this edge

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

Status as defined by [swTrackingIDError_e](#)

Remarks

Before calling this method, call [IEdge::GetTrackingIDsCount](#) to determine Count.

See [Tracking IDs](#) for more information.

See Also

[IEdge Interface](#)

[IEdge Members](#)

[IEdge::GetTrackingIDs Method](#)

[IEdge::RemoveTrackingID Method](#)

[IEdge::SetTrackingID Method](#)

[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetTwoAdjacentFaces Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IGetTwoAdjacentFaces Method (IEdge)

Obsolete. Superseded by [IEdge::IGetTwoAdjacentFaces2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetTwoAdjacentFaces( _  
    ByRef Face1 As Face, _  
    ByRef Face2 As Face _  
)
```

Visual Basic (Usage)

```
Dim instance As IEdge  
Dim Face1 As Face  
Dim Face2 As Face
```

```
instance.IGetTwoAdjacentFaces(Face1, Face2)
```

C#

```
void IGetTwoAdjacentFaces(  
    out Face Face1,
```



```
    out Face Face2
)
```

C++/CLI

```
void IGetTwoAdjacentFaces(
&    [Out] Face^ Face1,
&    [Out] Face^ Face2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Face1

Face2

v

Visual Basic for Applications (VBA) Syntax

See [Edge::IGetTwoAdjacentFaces](#).

See Also

[IEdge Interface](#)

[IEdge Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetTwoAdjacentFaces2 Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IGetTwoAdjacentFaces2 Method (IEdge)

Gets the two faces adjacent to an edge.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetTwoAdjacentFaces2( _
    ByRef Face1 As Face2, _
    ByRef Face2 As Face2 _
)
```

Visual Basic (Usage)



```
Dim instance As IEdge
Dim Face1 As Face2
Dim Face2 As Face2
```

```
instance.IGetTwoAdjacentFaces2(Face1, Face2)
```

C#

```
void IGetTwoAdjacentFaces2(
    out Face2 Face1,
    out Face2 Face2
)
```

C++/CLI

```
void IGetTwoAdjacentFaces2(
&    [Out] Face2^ Face1,
&    [Out] Face2^ Face2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Face1

Pointer to the first adjacent [face](#)

Face2
Pointer to the second adjacent [face](#)

Visual Basic for Applications (VBA) Syntax

See [Edge::IGetTwoAdjacentFaces2](#).

Remarks

This method is designed to be used with body-related edges, not reference curve or reference sketch edges. This method supports both solid and sheet bodies.

If...	Then the values returned are...
Two valid faces exist	Two faces
Only one valid face exists	One face and NULL

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetTwoAdjacentFaces2 Method](#)
[IVertex::GetAdjacentFaces Method](#)
[IVertex::IGetAdjacentFaces Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsParamReversed Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IsParamReversed Method (IEdge)

Gets whether the edge and its underlying curve have the same parameterization or if the direction is reversed.

.NET Syntax

Visual Basic (Declaration)

```
Function IsParamReversed() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Boolean  
  
value = instance.IsParamReversed()
```

C#

```
System.bool IsParamReversed()
```

C++/CLI

```
System.bool IsParamReversed();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the curve direction and edge direction are different, false if not

Visual Basic for Applications (VBA) Syntax

See [Edge::IsParamReversed](#).

Remarks

If this method returns True, then the curve direction and edge direction are different. As the parameter increases, the corresponding point on the edge moves from the end of the edge to the start.

If this method returns false, then the curve direction and edge direction are the same.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetCurveParams2 Method](#)
[IEdge::!GetCurveParams2 Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsTolerant Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : IsTolerant Method (IEdge)

Gets whether an edge is tolerant and its tolerance value.

.NET Syntax

Visual Basic (Declaration)

```
Function IsTolerant( _  
    ByRef Tolerance As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim Tolerance As System.Double  
Dim value As System.Boolean  
  
value = instance.IsTolerant(Tolerance)
```

C#

```
System.bool IsTolerant(  
    out System.double Tolerance  
)
```

C++/CLI

```
System.bool IsTolerant(  
    & [Out] System.double Tolerance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Tolerance
Edge tolerance or gap in meters

Return Value

True if tolerant, false if not tolerant

Visual Basic for Applications (VBA) Syntax

See [Edge::IsTolerant](#)

Example

[Get Maximum Edge and Vertex Gaps \(VBA\)](#)
[Get Maximum Edge and Vertex Gaps \(VB.NET\)](#)
[Get Maximum Edge and Vertex Gaps \(C#\)](#)

Remarks

If the tolerance (or gap) between common edges in a part:

- is greater than 1/2 of the session precision (i.e., > 5.0E-9 mm), then each edge is tolerant, and this method returns true.
- is less than or equal to 1/2 of the session precision (i.e., <= 5.0E-9 mm), then each edge is exact and not tolerant, and this method returns false.

Traverse part body edges and use this method to find the maximum edge gap (tolerance) in a part, which can also be found using the **Tools > Check** dialog in the SOLIDWORKS Design user interface.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IVertex::IsTolerant](#)

Availability

SOLIDWORKS 2009 SP2, Revision Number 17.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveId Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : RemoveId Method (IEdge)

Removes the edge ID assigned to this edge of an imported body.

.NET Syntax

Visual Basic (Declaration)

```
Sub RemoveId()
```

Visual Basic (Usage)



```
Dim instance As IEdge
```

```
instance.RemoveId()
```

C#

```
void RemoveId()
```

C++/CLI

```
void RemoveId();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Edge::RemoveId](#).

Remarks

SOLIDWORKS uses an edge ID to track edges of imported bodies.

The ID of the edge of an imported body:

- is not saved with the document.
- must be unique.
- can be changed by any third-party application. The intent is that if you [assign an ID to an edge of an imported body](#), you can refer to that edge within your application.
- is not a [persistent reference ID](#) or [tracking ID](#).

Use the [IAttribute](#) object to store data with an edge.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetID Method](#)
[IEdge::SetId Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveRedundantTopology Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : RemoveRedundantTopology Method (IEdge)

Removes redundant topology from the edge.

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveRedundantTopology() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim value As System.Boolean
```

```
value = instance.RemoveRedundantTopology()
```

C#

```
System.bool RemoveRedundantTopology()
```

C++/CLI

```
System.bool RemoveRedundantTopology();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if redundant topology was removed successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [Edge::RemoveRedundantTopology](#).

See Also

[IEdge Interface](#)
[IEdge Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveTrackingID Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : RemoveTrackingID Method (IEdge)

Removes a [tracking ID assigned to this edge](#).

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveTrackingID(  
    ByVal TrackingCookie As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim TrackingCookie As System.Integer  
Dim value As System.Integer  
  
value = instance.RemoveTrackingID(TrackingCookie)
```

C#

```
System.int RemoveTrackingID(  
    System.int TrackingCookie  
)
```

C++/CLI

```
System.int RemoveTrackingID(  
    & System.int TrackingCookie  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie
Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

Return Value

Status as defined by [swTrackingIDError_e](#)

Visual Basic for Applications (VBA) Syntax

See [Edge::RemoveTrackingID](#).

Example

[Get, Set, and Remove Tracking IDs on Edge \(VBA\)](#)

Remarks

See [Tracking IDs](#) for more information.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::SetTrackingID Method](#)
[IEdge::GetTrackingIDs Method](#)
[IEdge::GetTrackingIDsCount Method](#)
[IEdge::IGetTrackingIDs Method](#)
[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetId Method (IEdge)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : SetId Method (IEdge)

Sets the edge ID of this edge of an imported body.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetId(  
    ByVal IdIn As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim IdIn As System.Integer  
  
instance.SetId(IdIn)
```

C#

```
void SetId(  
    System.int IdIn  
)
```

C++/CLI

```
void SetId(  
& System.int IdIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IdIn

Edge ID of this edge of an imported body

Visual Basic for Applications (VBA) Syntax

See [Edge::SetId](#).

Remarks

SOLIDWORKS uses this ID to track edges of imported bodies.

The ID of the edge of an imported body:

- is not saved with the document.
- must be unique.
- can be changed by any third-party application. The intent is that if you [assign an ID to an edge of an imported body](#), you can refer to that edge within your application.
- is not the same as a [persistent reference ID](#) or [tracking ID](#).

Use the [IAttribute](#) object to store data with an edge.

See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::RemoveId Method](#)
[IEdge::GetID Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetTrackingID Method (IEdge)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEdge Interface](#) : SetTrackingID Method (IEdge)

Assigns a tracking ID to this edge.

.NET Syntax

Visual Basic (Declaration)

```
Function SetTrackingID( _  
    ByVal TrackingCookie As System.Integer, _  
    ByVal TrackingID As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEdge  
Dim TrackingCookie As System.Integer  
Dim TrackingID As System.Integer  
Dim value As System.Integer  
  
value = instance.SetTrackingID(TrackingCookie, TrackingID)
```

C#

```
System.int SetTrackingID(  
    System.int TrackingCookie,  
    System.int TrackingID  
)
```

C++/CLI

```
System.int SetTrackingID(  
& System.int TrackingCookie,  
& System.int TrackingID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie
Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)
TrackingID
Application-defined value for the tracking ID

Value	Action
>0	Edge is tracked
0	Tracking of edge is stopped

Return Value

Status as defined by [swTrackingIDError_e](#)

 Visual Basic for Applications (VBA) Syntax

See [Edge::SetTrackingID](#).

 Example

[Get, Set, and Remove Tracking IDs on Edge \(VBA\)](#)

 Remarks

You can assign tracking IDs to [bodies](#), edges, [faces](#), [loops](#), and [vertices](#) in parts and assemblies only; you cannot assign tracking IDs to these entities in drawings. See [Tracking IDs](#) for more information.

 See Also

[IEdge Interface](#)
[IEdge Members](#)
[IEdge::GetTrackingIDs Method](#)
[IEdge::RemoveTrackingID Method](#)
[IEdge::IGetTrackingIDs Method](#)
[IEdge::GetTrackingIDsCount Method](#)
[IModelDocExtension::FindTrackedObjects Method](#)

 Availability

